

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Examination - Summer 2024

Course: B. Tech. Branch: Electronics and Communication Engineering /

Electronics and Telecommunication Engineering Semester: IV

Subject Code & Name: Network Theory (BTETC401)

Max Marks: 60

Date: 12/06/2024

Duration: 3 Hr.

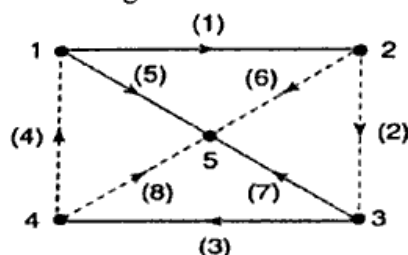
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

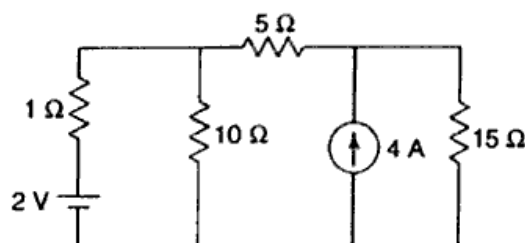
	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12

- | | | |
|---|-----|---|
| A) Define the following (i) Tree and co-tree (ii) Planar and non-Planar Graph | CO1 | 6 |
|---|-----|---|

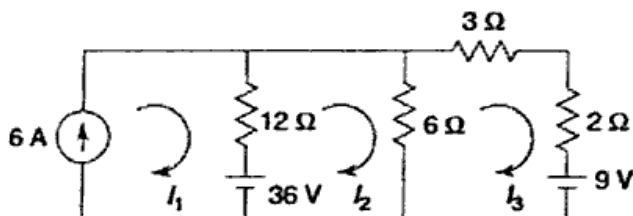
And for the graph shown in Fig. write the incidence matrix



- | | | |
|---|-----|---|
| B) State Norton's theorem. Find the current through the $10\ \Omega$ resistor using Norton's theorem. | CO1 | 6 |
|---|-----|---|



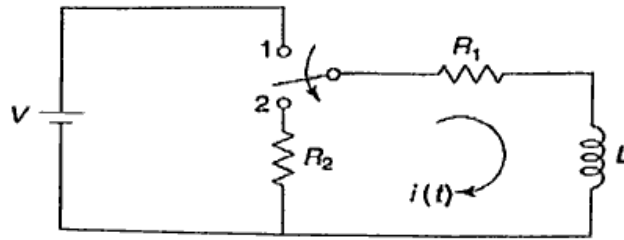
- | | | |
|---|-----|---|
| C) State and explain Kirchhoff's laws? Find the current through the $2\ \Omega$ resistor in the network of Fig shown below using mesh analysis. | CO1 | 6 |
|---|-----|---|



Q.2 Solve Any Two of the following.	12
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- | | | |
|--|----------|---|
| A) Derive an Expression for DC Response of an R-C Circuit | CO1, CO3 | 6 |
| B) Explain the initial conditions for Resistor, capacitor and inductor. | CO1, CO3 | 6 |
| C) In the network of Fig. shown, the switch is initially at the position 1. On the steady state having reached, the switch is changed to the position 2. | CO1, CO3 | 6 |

Find current $i(t)$.



Q. 3 Solve Any Two of the following.

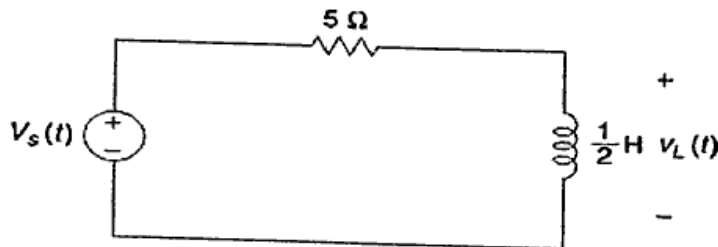
12

A) Explain the Behavior of basic elements in Laplace Transform and Find Laplace Transform of any two standard time domain functions **CO1, CO3**

6

B) Determine the expression for $V_L(t)$ in the network shown. Find the $V_L(t)$ when (i) $V_s(t) = \delta(t)$, and (ii) $V_s(t) = e^{-t}u(t)$. **CO1, CO3**

6



C) To find Initial and Final values if they exist of the signal with L.T

CO1

6

$$\frac{s+10}{s^2+3s+2} \text{ and } \frac{s^2+5s+2}{s^2+3s+2}$$

Q.4 Solve Any Two of the following.

12

A) Currents i_1 and i_2 entering at port 1 and port 2 respectively of a two-port network are given by **CO1, CO4**

6

$$i_1 = 0.5 V_1 - 0.2 V_2$$

$$i_2 = -0.2 V_1 + V_2$$

Find Y parameters. From Y parameters, check whether the network is reciprocal and symmetrical.

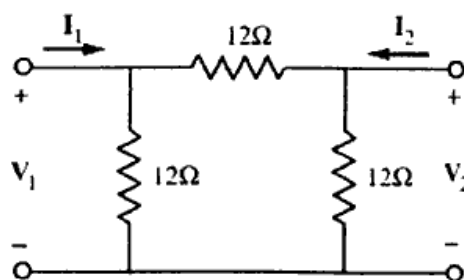
B) Explain the series and parallel connection of two port networks. Derive the interrelationship between transmission and hybrid two port network parameters. **CO1, CO4**

6

C) Define the terms: Driving-point Impedance Function, Driving-point Admittance Function. **CO1, CO4**

6

Find Z-parameter for the network given below



- Q. 5 Solve Any Two of the following.** **12**
- A) Realise Foster forms of the following function **CO1, CO3** **6**
- $$Z(s) = \frac{4(S^2 + 1)(S^2 + 16)}{S(S^2 + 4)}$$
- B) Test whether the polynomial **CO1, CO3** **6**
- $$P(s) = S^6 + 2S^5 + 8S^4 + 12S^3 + 20S^2 + 16S + 16$$
- is Hurwitz.
- C) Explain types of filters. Draw the characteristics of Low-pass filter and High-pass filters **CO1, CO3** **6**

***** End *****